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Trans-interface engineering for reliable perovskite solar cells with reproducible stability

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High reproducibility in interface-dominated device stability is crucial for the industrialization of perovskite solar cells. Conventional interface engineering uses planar interlayers to passivate and interlock the perovskite and the charge-transport layer, which often involves a trade-off between efficient charge transport and high stability. In this work, we use trans-interface engineering to create vertically aligned, mechanically tough nanolinks that penetrate both the perovskite and the charge-transport layer, simultaneously enhancing interface reliability and charge transfer. Here, as a proof of concept, we pre-incorporated triple and single organic linkers into the perovskite and the charge-transport layer, respectively, and then applied azobisisobutyronitrile-derived cyanoisopropyl radicals to trigger polymerization of the two linkers within their respective layers and to vertically bond them at the heterointerfacial contacts. A perovskite/charge-transport-layer interface enhanced by trans-interface engineering shows superior mechanical reliability compared with other reported interfacial approaches. The resultant perovskite solar cells deliver improved power conversion efficiencies of up to 27.04% (certified at 26.67%). Under maximum-power-point tracking at one-sun-intensity illumination, 92% of the tested perovskite solar cells reached T_{80} lifetimes of 1,400 h, demonstrating improved stability and reproducibility. This high reliability stems largely from the trans-interface engineering-induced suppression of mechanical failure in device operation. This work points to a new direction for improving perovskite solar cells using trans-interfacial materials, which is also applicable to other multilayered optoelectronic devices beyond photovoltaics.

Perovskite solar cells (PSCs) are thin-film photovoltaic devices with an active perovskite layer sandwiched between electron- and hole-transporting contacts. Despite the rapid increase in efficiency[1–8], stability remains a hurdle for commercialization[9–11]. The degradation of PSCs is known to be dominated by device interfaces, where two different mechanistic factors—chemical degradation and mechanical failure—are simultaneously in play[12–14]. Chemical degradation is generally associated with moisture, light or heat-triggered phase transitions or decomposition reactions of perovskite, which are temperature-dependent and usually cause a gradual decrease in the power conversion efficiency (PCE) during long-term operation[15–21]. Such degradation is often considered to be predictable based on thermodynamic analyses[22,23]. By contrast, mechanical failures are often caused by void formation or crack-induced interfacial delamination, causing a loss of the effective device area, especially when devices are subject to severe environmental changes (temperature swings and diurnal

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cycles)[24–28]. The mechanical failure of PSCs is often sudden and random, leading to unpredictable, steep PCE drops during device operation[29]. This uncertainty undermines our confidence in reliably assessing the commercialization potential of PSCs.

Many studies have largely focused on slowing chemical-related degradation pathways, and they use a single or a few devices to demonstrate the value of degradation-mitigation strategies[9,30–35]. The statistical reproducibility of device degradation is rarely considered. Mitigating mechanical failure in PSCs is a critical step towards greater confidence in achieving more reproducible, predictable device stability. Interface engineering is considered one of the most effective strategies for simultaneously enhancing the mechanical and chemical stability of PSCs. Conventional methods typically involve inserting a planar interlayer between the perovskite and the charge-transport layer (CTL)[36–38] (Supplementary Fig. 1). These approaches yield limited enhancement in interfacial mechanical adhesion, as they rely primarily on weak interactions, such as coordination or hydrogen bonding. Recent advances have extended the planar interlayers into the perovskite grain boundaries (GBs)[39–41] (Supplementary Fig. 1), but they do not extend into the CTL layer. In this context, incorporating toughening agents that root into the intra-perovskite, intra-CTL and perovskite/CTL interfaces is expected to holistically improve the reliability and optoelectronic properties of these interfaces.

In this work, we introduce trans-interface engineering (TIE), which establishes vertically aligned, mechanically tough nanolinks that root into both the perovskite and CTL layers. For a proof-of-concept demonstration, we pre-incorporated two cross-linkable monomers—triple linker trimethylolpropane triacrylate (TMTA) and single linker acrylamide (AM)—into the perovskite and the electron-transport layer (ETL), respectively. Then, we applied azobisisobutyronitrile (AIBN)-derived cyanoisopropyl radicals to induce the TMTA and AM to polymerize into PTMTA and PAM blocks that propagate along the intralayer interfaces of the perovskite and ETL, respectively, and eventually link at the perovskite/ETL heterointerface at a grain-level resolution. TIE doubles the interfacial toughness due to its unprecedented trans-interface PTMTA/PAM links. The TMTA and AM linkers can also passivate defects within and at the interface of the perovskite and ETL, thereby minimizing non-radiative recombination and improving the optoelectronic properties. As a result, PSCs deliver an improved PCE of 27.04% with a high open-circuit voltage V_{OC} = 1.220 V. More importantly, we performed a statistical analysis of the operational stability of PSCs based on over 50 individual devices by tracking the maximum power point (MPP) under one-sun-intensity illumination. We observed much higher reproducibility in TIE-based PSCs, which we attribute to reduced mechanical failure during long-term stability tests. We also demonstrated that TIE is generally applicable across a range of key device structures in PSCs, representing a paradigm shift in interface engineering research.

TIE synthesis and characterization

We employed the typical triple linker TMTA (Supplementary Fig. 2) for its high cross-linking density and strong resistance to water after polymerization[42,43]. We used the single linker AM (Supplementary Fig. 2) as the other monomer for its high hydrophilicity, enabling effective incorporation into metal-oxide-based CTLs[44]. When a mixed solution of the two monomers and AIBN initiator was heated at 50 °C under a nitrogen atmosphere, we observed that both can undergo self-polymerization to form PTMTA and PAM and also co-polymerization to form $P(AM)_m(TMTA)_n$ (Fig. 1a and Supplementary Figs. 3 and 4). The AIBN initiator thermally decomposes to generate cyanoisopropyl free radicals (Supplementary Fig. 5). These radicals interact with the C=C bonds in AM and TMTA, initiating chain propagation and triggering the formation of PAM, PTMTA and $P(AM)_m(TMTA)_n$. To confirm the co-polymerization, we dissolved TMTA and AIBN in *N*,*N*-dimethylformamide (DMF) and then added the solution to the as-polymerized PAM (Supplementary Fig. 6). After a heat treatment in N_2, we observed the formation of a bisegmented rod structure consisting of a transparent PTMTA segment chemically linked to white PAM. Interestingly, this structure consistently fractured within the PAM segment under a tensile force rather than at the PTMTA/PAM interface (Supplementary Fig. 6). This result reflects the co-polymerization at the PTMTA/PAM interface with high interfacial mechanical reliability.

We then applied TIE to n–i–p structured PSCs by pre-incorporating TMTA and AM into the perovskite precursor solution and the SnO_2 colloidal solution, respectively. Triggered by the AIBN-induced cyanoisopropyl radicals, TMTA and AM self-polymerize along perovskite GBs and SnO_2 interparticle boundaries, respectively, and eventually copolymerize at the perovskite/SnO_2 ETL interface (Fig. 1a). We employed transmission electron microscopy (TEM) to examine the interface cross section, which revealed a polymer network at the GBs of the perovskite layer and extending into the interparticle boundaries of the SnO_2 ETL (Supplementary Fig. 7). As shown in elemental mapping using high-angle annular dark-field scanning TEM (HAADF-STEM) (Fig. 1b), the polymer network, which traces from the perovskite GBs to the SnO_2 ETL and is amorphous, was clearly visualized by carbon (C) elemental mapping. This amorphous character was consistently observed across several representative regions with distinct GB geometries (Supplementary Fig. 7), confirming the general formation of the polymer network, irrespective of the local interfacial structure. More TEM evidence is provided in Supplementary Fig. 8. Then, we conducted a DMF rinsing test on the perovskite film. We found that in a TIE sample, this polymer network can remain adhered to the SnO_2 ETL (Supplementary Figs. 9 and 10). By contrast, when polymerization occurs only within the perovskite layer or only within the SnO_2 layer, no polymer network remains on the ETL after DMF rinsing. Also, we prepared a delaminated perovskite film sample with TIE and performed Fourier transform infrared (FTIR) spectroscopy (Supplementary Fig. 11). Typically, pure TMTA and AM exhibit characteristic C=O peaks at ~1,730 cm^{-1} and ~1,655 cm^{-1} (Supplementary Fig. 12)[42–44], both of which are well retained after polymerization (Fig. 1c). We identified the coexistence of the two characteristic C=O peaks on the bottom side of a flipped $CsPbI_2Br$ perovskite film, and importantly, both were fully retained even after prolonged water immersion (Fig. 1d). We selected $CsPbI_2Br$ perovskite to eliminate potential FTIR signal interference between FA^+ and the polymers. Because AM and PAM are water-soluble whereas $P(AM)_m(TMTA)_n$ is not, our observation indicates that co-polymerization between AM and TMTA occurred (Supplementary Fig. 13). Further spectroscopic evidence for the establishment of TIE is included in Supplementary Figs. 14 and 15. Collectively, these results demonstrate that TIE creates a chemically linked, solvent-resistant polymer network across the perovskite/ETL interface. Moreover, the versatility of TIE is highlighted by its successful extension to other polymerizable monomer pairs (Supplementary Fig. 16).

TIE-enhanced interfacial mechanical properties, charge dynamics and stability

We then investigated the mechanical behaviour of the TIE perovskite film. For comparison, we fabricated a control film with TMTA polymerization only in the perovskite layer, such that the polymer links did not cross the perovskite/ETL interface. We first used (quasi-)tape tests to evaluate the mechanical adhesion of both perovskite films[25,26]. As schematically illustrated in Fig. 2a, we evaluated the mechanical properties for the perovskite films that remained on the SnO_2 after peeling off the film. Based on 5 × 5 cross-cut film specimens (Supplementary Fig. 17), the mean adhesion quality improved from 1B (region affected by delamination between 35% and 65%) to 3B (region affected by delamination between 0% and 15%), according to the ASTM D3359 standard (Fig. 2b and Supplementary Table 1). We used scanning electron microscopy and atomic force microscopy to examine the exposed film bottoms. We observed for the TIE film that the detached regions were either covered by the SnO_2 ETL or contained numerous grain voids, which are

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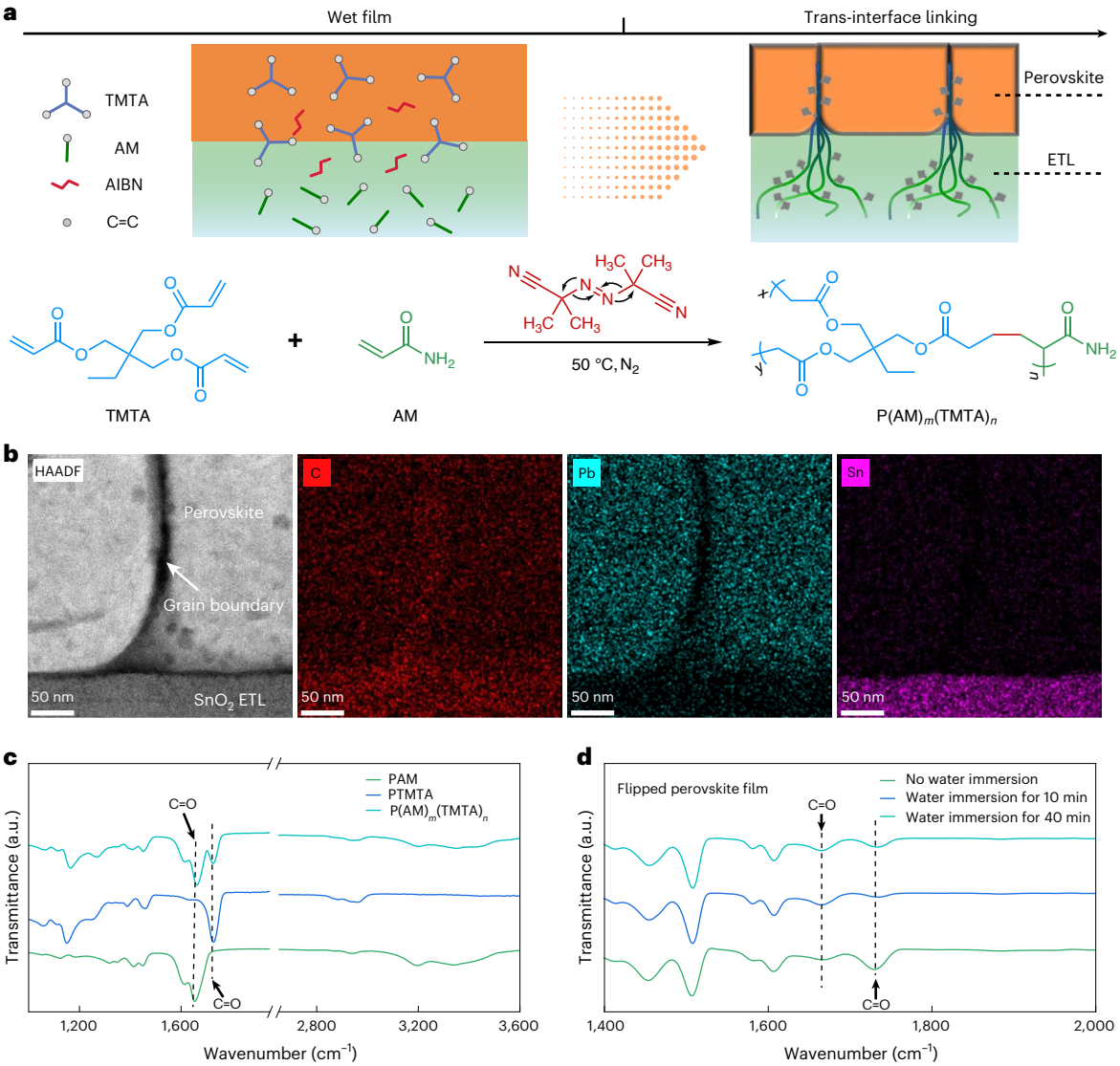


Fig. 1 | TIE synthesis and characterization. a, Schematic illustration of TIE with grain resolution across the entire ETL/perovskite heterointerface of PSCs (top). Schematic illustration of the polymerization paths of AM and TMTA induced by the AIBN initiator under heat treatment in a N_2 atmosphere (bottom). **b**, Cross-sectional HAADF image and corresponding elemental mappings (C, Pb and Sn) for a TIE-based PSC with a perovskite composition of $FA_{0.9}Cs_{0.1}PbI_3$. Note that, to more clearly visualize the interface, SnO_2 was deposited at three times the thickness (~100 nm) of the regular device (~30 nm). **c**, FTIR spectra for polymerized PAM, PTMTA and co-polymerized $P(AM)_m(TMTA)_n$. **d**, FTIR spectra of the flipped $CsPbI_2Br$ perovskite film with TIE after water immersion for different durations.

considered traces of intralayer fracture within the SnO_2 ETL or perovskite (Fig. 2c and Supplementary Figs. 18 and 19). This is attributed to the covalent bonding within the TIE links, which can boost the interfacial reliability between the perovskite and SnO_2 ETL layers and even surpass inter-perovskite-grain or inter-SnO_2-particle interactions. By contrast, such fractures rarely occurred in the control film.

Then, we conducted a nanoscratch test to provide a qualitative assessment of perovskite/SnO_2 adhesion[45] (Supplementary Fig. 20). The average critical load L_c increased from 2,239 μN (control) to 2,759 μN (TIE), demonstrating enhanced interfacial adhesion in the TIE sample. We further quantified the ETL/perovskite interfacial strength (maximum detachment stress) using tensile testing[25,26]. The strain–stress curves show that the strength of the sample with TIE (σ_{TIE} = 7.0 MPa) was twice that of the control samples ($\sigma_{control}$ = 3.5 MPa) (Supplementary Fig. 21). The statistical analysis of six samples of each type revealed that TIE increased the mean interfacial strength threefold, demonstrating a robust heterointerface with greater tensile resistance (Fig. 2d). Moreover, we measured the adhesion toughness G_c using the double-cantilever-beam delamination method[24,46,47]. For comparison, we prepared interfacial-link-based samples (with a PAM interlayer between the perovskite and SnO_2 layers) and control samples (with PTMTA polymerized in the perovskite) (see schematic illustration in Fig. 2e). As shown in Fig. 2e, the determined G_c of the SnO_2/perovskite interface increased threefold with TIE, from 0.67 ± 0.21 J m^{-2} (interfacial link) and 0.76 ± 0.22 J m^{-2} (control) to 1.72 ± 0.28 J m^{-2} (TIE). This significant improvement highlights that the covalently bonded and trans-interface polymer network established by TIE creates fundamentally superior adhesion, whereas the interfacial-link and control samples relied primarily on a single layer of intermolecular interactions (coordination bonds or hydrogen bonds) at the interface, which are substantially weaker than covalent bonds. Moreover, the TIE-induced mechanical reinforcement is reproducible across different chemically distinct polymer systems as well as different perovskite compositions (Supplementary Fig. 22 and Supplementary Note 1).

Notably, TIE preserved the intrinsic properties of both the perovskite and ETL layers. A detailed characterization revealed no significant

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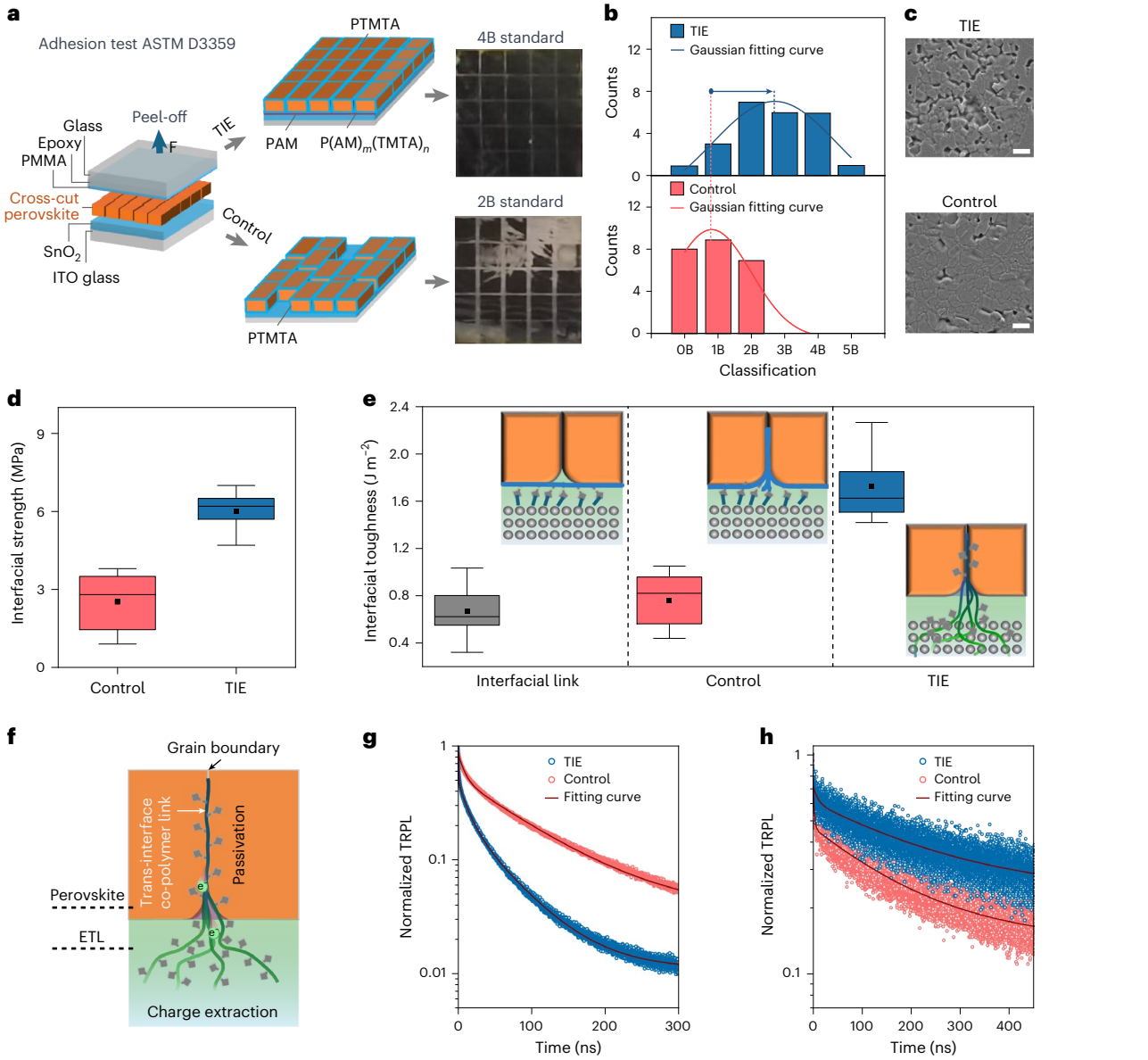


Fig. 2 | TIE-enhanced interfacial mechanical properties and charge dynamics. **a**, Schematic illustration of adhesion test and adhesion classification statistics for different perovskite films. **b**, Classification of adhesion and delamination results for different perovskite films (24 cross-cut sections for each film type). **c**, Scanning electron microscopy images of the perovskite regions that detached from the upper glass substrate. **d**, Interfacial strength statistics at the SnO_2/perovskite interface for different perovskite films (sample size $n = 6$ for both cases). **e**, Interfacial toughness of the SnO_2/perovskite interface for different interface designs (interfacial link, control and TIE). Data are presented as mean ± standard deviation (sample size $n = 10$ for all cases). Insets: schematic illustrations of the corresponding interface structures. The whiskers in **d** and **e** represent maxima and minima. The bounds of the boxes are the 25th and 75th percentiles. The centre is the mean. **f**, Schematic of grain-resolved TIE links, which passivate defects and enable charge transfer across the ETL/perovskite heterointerface. **g**, TRPL spectra of perovskite films deposited on SnO_2 ETLs with and without TIE. **h**, TRPL spectra of control and TIE films detached from SnO_2 ETL-coated substrates. The perovskite film composition here is $FA_{0.9}Cs_{0.1}PbI_3$. Scale bars, 1 μm (**c**).

alterations in perovskite grain size, surface morphology, crystallinity, orientation, bandgap or optical transmittance compared with control films (Supplementary Figs. 23, 24a and 25). Despite this structural and optical neutrality, the TIE link can play a positive electronic role at the interface. As illustrated in Fig. 2f, such molecular bridging simultaneously passivates defects at perovskite GBs and ETL interparticle boundaries via Lewis acid–base interactions and hydrogen bonding (Supplementary Figs. 24b and 26), while creating an uninterrupted charge transport network across the entire ETL/perovskite heterojunction. This unique combination of interface preservation and electronic modulation enables efficient electron extraction without compromising the intrinsic optoelectronic properties of either layer. To investigate the interfacial charge dynamics, we conducted steady-state photoluminescence (PL) and time-resolved PL (TRPL) measurements. With TIE, a more effective PL quenching was observed (Fig. 2g and Supplementary Fig. 27). The average PL lifetime dropped three times for the TIE film compared with the control film (Supplementary Table 2). These findings confirm enhanced charge extraction at the perovskite/ETL heterointerface. We also evaluated the electronic quality of perovskite films detached from SnO_2 ETL substrates using PL and TRPL measurements (Supplementary Fig. 28). The TIE sample shows an increase in PL intensity and carrier lifetime compared with the control sample (Fig. 2h, Supplementary Note 2 and Supplementary Table 2). We measured the evolution of the space-charge-limited current to

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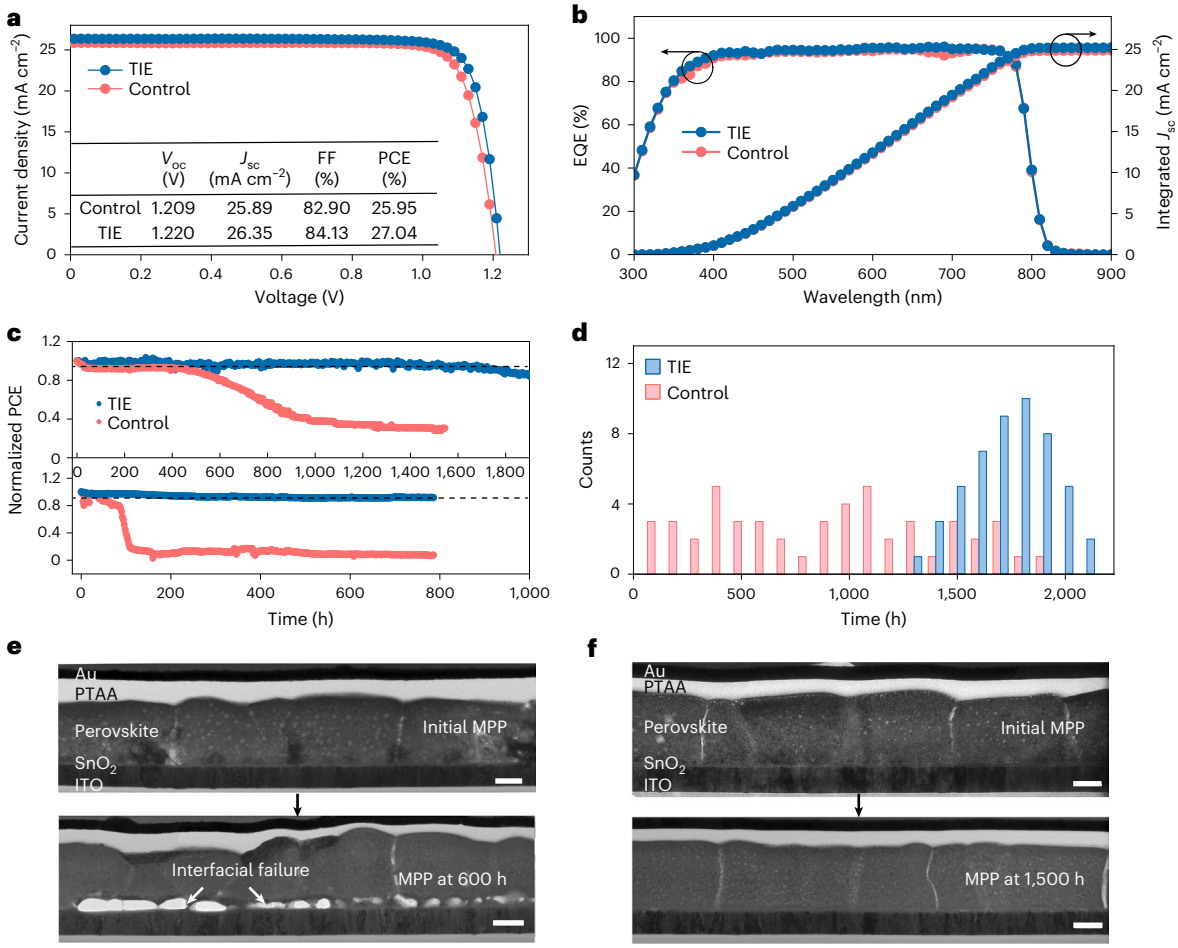


Fig. 3 | TIE-enhanced photovoltaic performance and durability of PSCs. a, *J*–*V* curves and photovoltaic parameters of champion PSCs under reverse scanning. **b**, EQE and integrated current density curves for the control (24.88 mA cm^{-2}) and TIE (25.41 mA cm^{-2}) devices. **c**, Operational stability measured by MPP tracking under one-sun-intensity illumination in N_2 at RT and 85 °C for control and TIE devices. The initial PCEs of control and TIE devices in **c** were 22.82% and 23.73%, respectively. **d**, Histogram showing T_{80} lifetime of MPP operational stability for different PSCs (sample size $n = 50$ for both cases). **e**, Cross-sectional TEM images of control devices before and after MPP tracking at RT. **f**, Cross-sectional TEM images of TIE devices before and after MPP. The devices in **a** and **b** have a configuration of glass/ITO/SnO_2/perovskite $(FA_{0.95}Cs_{0.05}PbI_3)_{0.975}(MAPbBr_3)_{0.025}$/spiro-OMeTAD/Au and were used to assess the efficiency potential of TIE PSCs. The devices in **c**–**f** were configured as glass/ITO/SnO_2/perovskite $(FA_{0.9}Cs_{0.1}PbI_3)$/PTAA/Au and used to assess the stability potential of TIE PSCs. Scale bars, 200 nm (**e**,**f**). FF, fill factor.

estimate the electronic defect density N_t. As shown in the dark current versus voltage curves of electron-only devices based on the TIE and control samples (Supplementary Fig. 29), the determined trap-filled voltages V_{TFL} were 0.177 V and 0.138 V, corresponding to trap densities of 5.76×10^{15} cm^{-3} and 4.48×10^{15} cm^{-3} for the control and TIE samples (Supplementary Table 3). We then determined the electron mobilities μ_e of the control and TIE films, which were 0.44 cm^2 V^{-1} s^{-1} and 0.77 cm^2 V^{-1} s^{-1}, respectively (Supplementary Fig. 30). Moreover, TIE increased the conductivity σ of the SnO_2 ETL from 7.2×10^{-3} S m^{-1} to 2.5×10^{-2} S m^{-1} (Supplementary Table 4). There is an extended relevant discussion in Supplementary Note 3. Furthermore, we compared the chemical stability of the TIE and control films under 25 °C and 85% relative humidity (Supplementary Figs. 31 and 32). As both TIE and the control samples contain PTMTA at perovskite GBs, the film stability improvement in the TIE sample could be attributed to a more intact interface microstructure, as elaborated above.

Enhanced PCEs and reproducible device stability of TIE-based PSCs

The n–i–p devices were used to evaluate the photovoltaic performance of PSCs. As shown in the current density versus voltage (*J*–*V*) curves in Fig. 3a, the TIE device with an optimal monomer concentration of 1 mg ml^{-1} (Supplementary Fig. 33 and Supplementary Note 4) achieved a champion PCE of 27.04% during a reverse scan (certified 26.67%; Supplementary Fig. 34), which is better than the control device with an efficiency of 25.95%. Also, when measured in reverse and forward scans, the TIE device exhibited negligible hysteresis (Supplementary Fig. 35). The resultant TIE device can exhibit better reproducibility than the control device, as indicated by the statistics for 50 individual devices under each condition (Supplementary Fig. 36). The average PCEs increased from 25.25% for control devices to 26.29% for TIE devices. We determined the bandgap E_g of different devices by differentiating their external quantum efficiency (EQE) spectra[48], which had an inflection point at 800 nm, corresponding to 1.55 eV (Supplementary Fig. 37). Therefore, the V_{OC} deficit ($E_g - qV_{OC}$) of the TIE-based device was calculated to be as low as 0.33 V (ref. 49), which is one of the lowest values reported so far for PSCs with $E_g \approx 1.55$ eV (Supplementary Fig. 38 and Supplementary Table 5). Additionally, the integrated current density J_{SC} from the EQE closely matched the value extracted from the *J*–*V* curve (Fig. 3b). The enhanced EQE in the wavelength ranges of 650 nm to 780 nm and 350 nm to 500 nm can be attributed to the optimized energy-level alignment at the device heterointerface (Supplementary Fig. 39 and Supplementary Note 5). The improved V_{OC} and suppressed hysteresis are attributed to enhanced

interfacial electronic and structural properties and charge dynamics enabled by the TIE links. We also confirmed that the initiator AIBN was largely removed during processing, with only trace amounts remaining in the final device, which had a negligible impact on device performance (Supplementary Figs. 40 and 41 and Supplementary Note 6). Furthermore, the TIE strategy was successfully extended to p–i–n device architectures, demonstrating its broad applicability (Supplementary Fig. 44).

We then conducted a series of durability tests for different PSC devices under various external stresses. We used poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA) as the hole-transporting layer in all PSCs subjected to durability testing. First, we performed MPP tracking tests under one-sun-intensity illumination at room temperature (RT) following the International Summit on Organic Photovoltaic Stability (ISOS) ISOS-L-1 protocol[50]. To illustrate the evolution of efficiency, two representative devices from each group were selected, as shown in Fig. 3c. The control device, with an initial PCE of 22.82%, remained stable for the initial 400 h but then experienced a relatively steep decline between 400 h and 600 h. Note that such performance degradation trend under MPP tracking was frequently observed in the tested control devices. By contrast, the TIE-based device demonstrated substantially enhanced stability, retaining 95% of its initial PCE (23.73%) after 1,500 h under the same conditions. Furthermore, we analysed the operational stability of 50 devices under each condition based on the time for device's PCE to drop to 80% of its initial value (T_{80} lifetime), presented in the form of histograms (Fig. 3d and Supplementary Figs. 45 and 46). The control devices exhibited a broad distribution of T_{80} lifetimes, ranging from only tens of hours to over 1,800 h. By contrast, the TIE-based devices showed a much narrower distribution, clustered around 1,800 h, indicating high reproducibility. Specifically, 92% of the tested TIE-based PSCs reached a T_{80} lifetime of 1,400 h under MPP tracking. We also evaluated the MPP operational stability under one-sun-intensity illumination at 85 °C using the more stringent ISOS-L-2 protocol (Fig. 3c and Supplementary Fig. 47). After 800 h of ageing, all four TIE-based devices had retained over 90% of their initial PCEs (~23%), further demonstrating high stability and reproducibility. By contrast, the control devices frequently showed signs of catastrophic failure within 200 h. Regardless of MPP tracking at RT or 85 °C, we attributed the rapid decline in efficiency of the control devices to mechanical failure at the perovskite/ETL heterointerface during the MPP operation (Fig. 3e and Supplementary Figs. 48 and 49). By contrast, TIE-treated devices not only demonstrated enhanced chemical stability but also improved mechanical reliability, leading to a considerable increase in overall durability (Fig. 3f and Supplementary Fig. 48). Furthermore, we measured the PCE of PSCs under thermal fatigue to assess their interfacial mechanical reliability. The thermal-cycling test was performed under the ISOS-T-3 protocol. For the TIE PSC, the initial PCE was almost retained after 300 temperature cycles, markedly outperforming the control devices (84%; Supplementary Fig. 50). In addition, we placed encapsulated devices in a chamber for a damp heat test (85 °C and 85% relative humidity) following the ISOS-D-3 protocol. The TIE-based devices maintained 90% of their initial PCEs over 1,000 h under damp heat ageing, whereas the control devices retained <50% of their initial PCEs under the same conditions (Supplementary Fig. 51). All these stability results demonstrate that the current TIE strategy significantly enhances the long-term durability and the stability reproducibility, making it a promising approach for improving device performance under harsh environmental conditions.

Conclusion and outlook

We have demonstrated an effective TIE strategy that mechanically toughens the perovskite/CTL interface while passivating perovskite and CTLs in PSCs. By leveraging TIE nanolinks, PSCs exhibit superior resistance to external stimuli, including heat, light and moisture, ensuring consistent mechanical integrity and reproducible stability. These heterointerfaces also enhanced the optoelectronic properties of the device through defect passivation, enabled by intralayer chemical interactions among the TIE nanolinks, the perovskite and the ETL. We note that the radical-derived co-polymer network may undergo long-term ageing, yet the present proof-of-concept TIE devices already withstand several accelerated-ageing protocols. Future integration of intrinsically more stable monomers and control over in situ polymerization within the device architecture could further extend operational durability. This work highlights a promising new approach to improving the stability and reproducibility of PSCs by using trans-interface linking materials. TIE adds a new dimension of control to established interface engineering of the device architecture. By strategically designing monomer chemistry and polymerization kinetics to engineer mechanically and electronically tailored polymers, TIE could be a robust foundation for next-generation photovoltaic and optoelectronic devices, delivering improved performance and reliability.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41566-026-02007-w.

References

1. Ugur, E. et al. Stable and uniform self-assembled organic diradical molecules for perovskite photovoltaics. *Science* **385**, 533–538 (2025).
2. You, S. et al. C_{60}-based ionic salt electron shuttle for high-performance inverted perovskite solar modules. *Science* **388**, 964–968 (2025).
3. Li, Q. et al. Graphene-polymer reinforcement of perovskite lattices for durable solar cells. *Science* **387**, 1069–1077 (2025).
4. Yang, Y. et al. Amidination of ligands for chemical and field-effect passivation stabilizes perovskite solar cells. *Science* **386**, 898–902 (2024).
5. Zhang, Y. et al. Nonalloyed α-phase formamidinium lead triiodide solar cells through iodine intercalation. *Science* **387**, 284–290 (2025).
6. Wang, H. et al. Impurity-healing interface engineering for efficient perovskite submodules. *Nature* **634**, 1091–1095 (2024).
7. Liu, J. et al. Perovskite/silicon tandem solar cells with bilayer interface passivation. *Nature* **635**, 596–603 (2024).
8. Best research-cell efficiency chart. *National Laboratory of the Rockies* https://www.nlr.gov/pv/cell-efficiency (2026).
9. Sun, X. et al. Vapor-assisted surface reconstruction enables outdoor-stable perovskite solar modules. *Science* **388**, 957–963 (2025).
10. Zhao, X. et al. Operationally stable perovskite solar modules enabled by vapor-phase fluoride treatment. *Science* **385**, 433–438 (2024).
11. Jiang, Q. et al. Towards linking lab and field lifetimes of perovskite solar cells. *Nature* **623**, 313–318 (2023).
12. Dai, Z. & Padture, N. P. Challenges and opportunities for the mechanical reliability of metal halide perovskites and photovoltaics. *Nat. Energy* **8**, 1319–1327 (2023).
13. Chen, J. et al. Determining the bonding–degradation trade-off at heterointerfaces for increased efficiency and stability of perovskite solar cells. *Nat. Energy* **10**, 181–190 (2025).
14. Zhou, Y., Herz, L. M., Jen, A. K.-Y. & Saliba, M. Advances and challenges in understanding the microscopic structure–property–performance relationship in perovskite solar cells. *Nat. Energy* **7**, 794–807 (2022).
15. Wang, Y. et al. Homogenized contact in all-perovskite tandems using tailored 2D perovskite. *Nature* **635**, 867–873 (2024).

16. Fei, C. et al. Strong-bonding hole-transport layers reduce ultraviolet degradation of perovskite solar cells. *Science* **384**, 1126–1134 (2024).
17. Tang, H. et al. Reinforcing self-assembly of hole transport molecules for stable inverted perovskite solar cells. *Science* **383**, 1236–1240 (2024).
18. Ye, L. et al. Superoxide radical derived metal-free spiro-OMeTAD for highly stable perovskite solar cells. *Nat. Commun.* **15**, 7889 (2024).
19. Frohna, K. et al. The impact of interfacial quality and nanoscale performance disorder on the stability of alloyed perovskite solar cells. *Nat. Energy* **10**, 66–76 (2025).
20. Li, G. et al. Highly efficient p-i-n perovskite solar cells that endure temperature variations. *Science* **379**, 399–403 (2023).
21. Shen, Y. et al. Strain regulation retards natural operation decay of perovskite solar cells. *Nature* **635**, 882–889 (2024).
22. Zhao, X. et al. Accelerated aging of all-inorganic, interface-stabilized perovskite solar cells. *Science* **377**, 307–310 (2022).
23. Burlingame, Q. C., Loo, Y.-L. & Katz, E. A. Accelerated ageing of organic and perovskite photovoltaics. *Nat. Energy* **8**, 1300–1302 (2023).
24. Dai, Z. et al. Interfacial toughening with self-assembled monolayers enhances perovskite solar cell reliability. *Science* **372**, 618–622 (2021).
25. Duan, T. et al. Chiral-structured heterointerfaces enable durable perovskite solar cells. *Science* **384**, 878–884 (2024).
26. Hao, M. et al. Flattening grain-boundary grooves for perovskite solar cells with high optomechanical reliability. *Adv. Mater.* **35**, 2211155 (2023).
27. Cheng, Q. et al. Bidirectional chemo-mechanical interface stabilization in perovskite solar cells. *J. Am. Chem. Soc.* **148**, 31146–31153 (2026).
28. Dong, Q. et al. Interpenetrating interfaces for efficient perovskite solar cells with high operational stability and mechanical robustness. *Nat. Commun.* **12**, 973 (2021).
29. Dong, Q. et al. Flexible perovskite solar cells with simultaneously improved efficiency, operational stability, and mechanical reliability. *Joule* **5**, 1587–1601 (2021).
30. Guo, P. et al. Interfacial embedding of laser-manufactured fluorinated gold clusters enabling stable perovskite solar cells with efficiency over 24%. *Adv. Mater.* **33**, 2101590 (2021).
31. Lin, X. et al. In situ growth of graphene on both sides of a Cu–Ni alloy electrode for perovskite solar cells with improved stability. *Nat. Energy* **7**, 520–527 (2022).
32. Luo, J. et al. Polymer-acid-metal quasi-ohmic contact for stable perovskite solar cells beyond a 20,000-hour extrapolated lifetime. *Nat. Commun.* **15**, 2002 (2024).
33. Shen, Z. et al. Efficient and stable perovskite solar cells with regulated depletion region. *Nat. Photonics* **18**, 450–457 (2024).
34. Paik, M. J., Kim, Y. Y., Kim, J., Park, J. & Seok, S. I. Ultrafine SnO_2 colloids with enhanced interface quality for high-efficiency perovskite solar cells. *Joule* **8**, 2073–2086 (2024).
35. Grancini, G. et al. One-year stable perovskite solar cells by 2D/3D interface engineering. *Nat. Commun.* **8**, 15684 (2017).
36. Zhu, X. et al. Restrictive heterointerfacial delamination in flexible perovskite photovoltaics using a bifacial linker. *Adv. Mater.* **37**, 2419329 (2025).
37. Yang, I. S. et al. Simultaneous enhancement of efficiency and operational-stability of mesoscopic perovskite solar cells via interfacial toughening. *Adv. Mater.* **36**, 2308819 (2024).
38. Cai, W. et al. Interlayer reinforcement for improved mechanical reliability for wearable perovskite solar cells. *Energy Environ. Sci.* **17**, 8162–8173 (2024).
39. Wang, M. et al. Uracil induced simultaneously strengthening grain boundaries and interfaces enables high-performance perovskite solar cells with superior operational stability. *Adv. Mater.* **36**, 2306415 (2024).
40. Li, Z. et al. In-situ cross-linked polymers for enhanced thermal cycling stability in flexible perovskite solar cells. *Angew. Chem. Int. Ed.* **64**, e202421063 (2025).
41. Li, Z. et al. Boosting mechanical durability under high humidity by bioinspired multisite polymer for high-efficiency flexible perovskite solar cells. *Nat. Commun.* **16**, 1771 (2025).
42. Li, X. et al. In-situ cross-linking strategy for efficient and operationally stable methylammoniun lead iodide solar cells. *Nat. Commun.* **9**, 3806 (2018).
43. Zhang, H. et al. Multifunctional crosslinking-enabled strain-regulating crystallization for stable, efficient α-$FAPbI_3$-based perovskite solar cells. *Adv. Mater.* **33**, 2008487 (2021).
44. Ge, C. et al. Thermal dynamic self-healing supramolecular dopant towards efficient and stable flexible perovskite solar cells. *Angew. Chem. Int. Ed.* **61**, e202116602 (2022).
45. Li, Z. et al. Hyperbranched polymer functionalized flexible perovskite solar cells with mechanical robustness and reduced lead leakage. *Nat. Commun.* **14**, 6451 (2023).
46. Tang, X. et al. Enhancing the efficiency and stability of perovskite solar cells via a polymer heterointerface bridge. *Nat. Photonics* **19**, 701–708 (2025).
47. Dai, Z. et al. Effect of grain size on the fracture behavior of organic-inorganic halide perovskite thin films for solar cells. *Scr. Mater.* **185**, 47–50 (2020).
48. Wu, S. et al. Redox mediator-stabilized wide-bandgap perovskites for monolithic perovskite-organic tandem solar cells. *Nat. Energy* **9**, 411–421 (2024).
49. Jeong, M. et al. Stable perovskite solar cells with efficiency exceeding 24.8% and 0.3-V voltage loss. *Science* **369**, 1615–1620 (2020).
50. Khenkin, M. V. et al. Consensus statement for stability assessment and reporting for perovskite photovoltaics based on ISOS procedures. *Nat. Energy* **5**, 35–49 (2020).

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Methods

Raw chemicals

Formamidinium iodide (FAI), methylammonium bromide, methylammonium chloride (MACl), 4-methoxy-phenethylammonium iodide (4-OMe-PEAI) and FK209-Co(III) bis(trifluoromethanesulfonyl) imide (TFSI) salt were purchased from Greatcell Solar Materials. DMF, dimethyl sulfoxide (DMSO), diethyl ether, isopropyl alcohol (IPA), ethyl acetate, 4-tert-butylpyridine, chlorobenzene (CB), acetonitrile, potassium chloride (KCl) and cadmium iodide (CdI_2) were purchased from Sigma-Aldrich. AIBN, AM, TMTA, ethylene glycol dimethacrylate, methyl methacrylate, Li-TFSI, tin tetrachloride ($SnCl_4$) and tetrahydrofuran (THF) were purchased from Macklin. Caesium iodide (CsI), lead bromide ($PbBr_2$), lead iodide (PbI_2), PTAA, phenyl-C61-butyric acid methyl ester ($PC_{61}BM$), C_{60}, bathocuproine and phenethylammonium iodide (PEAI), were purchased from Xi'an Yuri Solar Co., Ltd. Moreover, 4-isopropyl-4′-methyldiphenyliodonium tetrakis (pentafluorophenyl) borate (TPFB) was purchased from TCI, and 2,2′,7,7′-tetrakis[*N*,*N*-di(4-methoxyphenyl)amino]-9,9′-spirobifluorene (spiro-OMeTAD) was purchased from Borun Chemical Co., Ltd. The NiO_x powder was sourced from Advanced Election Technology Co., Ltd. Me-4PACz was acquired from Tokyo Chemical Industry. All materials were used without further purification.

Preparation of $MAPbBr_3$ single crystals

A 0.2 M precursor solution was prepared by dissolving MABr (22.4 mg) and $PbBr_2$ (73.4 mg) in 1 ml DMF. $MAPbBr_3$ single crystals were then prepared by vapour diffusion of an antisolvent (dichloromethane) into the solution at RT[51].

Preparation of SnO_2 colloidal solutions and SnO_2 ETLs

$SnCl_4$ (195 mg) and CdI_2 (8.2 mg) were added to ice water (15 ml) slowly under stirring. Then the solution was kept in an oven at 90 °C for 1.5 h to trigger the hydrolysis reaction. After the precursor had returned to RT, the SnO_2 nanoparticles were centrifuged and washed with ethanol. Finally, the SnO_2 nanoparticles were dispersed in ethanol via sonication to form a colloidal solution[52]. This SnO_2 colloidal solution was spin-coated onto the substrate at 3,000 rpm for 30 s and annealed at 100 °C for 5 min to form the SnO_2 ETLs. These ETLs were used for both the pristine and control devices. For the TIE device, 1 mg AM was added to 1 ml SnO_2 colloidal solution. Then the mixed SnO_2 colloidal solution was deposited onto the substrate through spin-coating at 3,000 rpm for 30 s and annealing at 100 °C for 5 min to form SnO_2 ETLs with AM (SnO_2-AM). At this stage, the AM was still in its monomeric state within the SnO_2 layer.

Preparation of $FA_{0.9}Cs_{0.1}PbI_3$ perovskite films with or without TIE treatment

For the fabrication of pristine $FA_{0.9}Cs_{0.1}PbI_3$ perovskite films, a 1.4 M precursor solution was prepared by dissolving PbI_2 (645.4 mg), FAI (216.7 mg), CsI (36.4 mg) and MACl (28.3 mg) in 1 ml DMF/DMSO mixed solvent (4:1, volume ratio (v/v)). The perovskite film was formed via a consecutive two-step spin-coating process at 1,000 rpm for 10 s and 4,000 rpm for 30 s. During the second step, 200 µl ethyl acetate was poured onto the substrate 20 s before programme completion. The substrate was then annealed on a hotplate at 160 °C for 10 min to form the $FA_{0.9}Cs_{0.1}PbI_3$ perovskite film. For the control and TIE samples, 1 mg TMTA and 0.1 mg AIBN were added to 1 ml perovskite precursor solution. For the control samples, the perovskite precursor solution was deposited onto an SnO_2-coated substrate. For the TIE samples, the perovskite precursor solution was deposited onto an SnO_2/AM-coated substrate. The subsequent formation of the film followed the identical spin-coating, antisolvent treatment and thermal annealing steps used for the pristine films.

Preparation of $CsPbI_2Br$ perovskite films with or without TIE treatment

For the fabrication of pristine $CsPbI_2Br$ perovskite films, a 1.2 M precursor solution was prepared by dissolving PbI_2 (276.6 mg), CsI (311.8 mg) and $PbBr_2$ (220.2 mg) in 1 ml DMF/DMSO mixed solvent (4:1, v/v). For the TIE device, 1 mg TMTA and 0.1 mg AIBN were added to 1 ml perovskite precursor solution. The perovskite film was formed via a consecutive spin-coating process at 4,000 rpm over 30 s. The substrate was then annealed on a hotplate at 50 °C for 2 min and 160 °C for 10 min to form the $CsPbI_2Br$ perovskite film.

Sample specimen preparation for FTIR measurements

The SnO_2 colloidal solution (with and without AM) was deposited onto the indium tin oxide (ITO) substrate through spin-coating at 3,000 rpm for 30 s, followed by annealing at 100 °C for 5 min. To prevent FAI from influencing the FTIR measurements, 1.2 M $CsPbI_2Br$ (with and without TMTA and AIBN) in DMF/DMSO mixed solvent (4:1, v/v) was deposited onto the ITO/SnO_2 substrate at a spin rate of 4,000 rpm over 30 s, followed by annealing at 50 °C for 2 min and 160 °C for 10 min. Then epoxy resin was applied to the perovskite layer to bond it to the glass substrate. Finally, the perovskite layer was peeled off the ITO/SnO_2 substrate, and the buried surface was tested. For the preparation of PAM and PTMTA, 150 mg of AM or TMTA was dissolved in 1 ml THF, and 3 mg AIBN was added as an initiator. For the corresponding co-polymer, 150 mg AM, 150 mg TMTA and 6 mg AIBN were dissolved in 1 ml THF. The solution was stirred at 70 °C for 24 h to initiate polymerization. For a different polymer system, we replaced TMTA with ethylene glycol dimethacrylate and methyl methacrylate.

Sample specimen preparation for matrix-assisted laser desorption/ionization time-of-flight mass spectrometry

Mass spectra were obtained using a mass spectrometer (Voyager Elite MALDI-TOF). For the preparation of the co-polymer, 1 mg AM, 1 mg TMTA and 0.05 mg AIBN were dissolved in 1 ml THF. The solution was stirred at 70 °C to cause polymerization. The sample was prepared by mixing a co-polymer solution, a 2,5-dihydroxybenzoic acid solution (50 mg ml^{-1} in THF) and a potassium trifluoroacetate solution (1 mg ml^{-1} in THF) in a 10:10:1 (v/v) ratio.

Sample specimen preparation for TEM characterization

A nanofabrication tool with a dual focused ion beam (FEI Helios G4 UX) was employed to prepare cross-sectional TEM samples of PSCs. Platinum (Pt) layers were deposited on the top surface of the devices through both electron and ion deposition for protection. Subsequently, rough milling (30 kV, 20 nA), fine milling (30 kV, 2.4 nA) and U-cut (30 kV, 2.4 nA) processes were performed to create the lamella. The lamella was then removed from the substrate using an easy lift. After removing the PSC, the TEM grid was placed into the focused-ion-beam chamber. The lamella was attached to the TEM grid and thinned to less than 100 nm using a gallium ion beam at a stable voltage of 30 kV and a current that was decreased from 750 pA to 41 pA. To mitigate damage from ion injection, a gallium ion beam with an accelerating voltage reduced to 2 kV and a working current of 44 pA was used to polish the sample surface. The prepared TEM samples were stored in a nitrogen glovebox before characterization.

Materials characterization

X-ray diffraction was performed with a high-resolution diffractometer with Cu Kα radiation (λ = 1.54 Å; Empyrean MultiCore PIX3D, Malvern Panalytical BV). Simultaneous thermogravimetry-differential scanning calorimetry was performed using a thermal analysis system (STA 2500, Netzsch). The steady-state PL and TRPL were measured using a 375-nm picosecond laser (PicoQuant LDH-D-C-375) as excitation. Top-view and cross-sectional scanning electron microscopy images were obtained using a field-emission scanning electron

microscope (JSM-6700F, JEOL). Atomic force microscopy images were obtained using a Bruker Dimension Icon instrument. Absorption and transmission were measured using an ultraviolet–visible spectrometer (Agilent). FTIR spectroscopy was measured using an infrared Fourier transform microscope (Vertex 70 Hyperion 1000, Bruker). Ultraviolet photoelectron spectroscopy was conducted on an Axis Supra+ (Kratos Analytical Limited). Nano-indentation and nanoscratch data were measured by a Hysitron TI 900. Adhesion toughness was measured by Delaminator (v10.0) 4-Pt (DTS). TEM characterization of the cross section of PSCs was carried out on an aberration-corrected microscope (Spectra 300, Thermo Fisher Scientific) equipped with dual spherical aberration correctors at an accelerating voltage of 300 kV. The probe convergence half-angle was set to 17.9 mrad, and the HAADF detector collected scattered electrons over an angular range from 49 mrad to 200 mrad. To mitigate electron-beam-induced irradiation damage, the electron probe current was tuned down to 1 pA, with a per-pixel dwell time restricted to 2 µs for imaging. For the energy-dispersive X-ray spectroscopy mapping acquisition, repeated scans were performed over the same sample region using a Super-X EDS detector, with the probe current fixed at 5 pA throughout data collection.

Mechanical testing

For the delamination tests of perovskite films, the samples were prepared with the structure glass substrate/ITO/SnO_2/perovskite/poly(methyl methacrylate) (PMMA)/epoxy resin/glass substrate through the following process. First, a PMMA solution (10 mg ml^{-1} in CB) was spread onto the perovskite films, followed by spin-coating at 6,000 rpm for 30 s. The function of the PMMA layer was to prevent the epoxy resin from intruding into the perovskite films. Then, the epoxy resin was applied to the PMMA layer to bond the glass substrate. Finally, the epoxy resin was cured for over 24 h. To measure the stress–displacement curves, the methodology detailed in the literature was employed using a tensile testing machine (E3000, Instron)[25,26]. The structure of the prepared samples (15 mm × 15 mm) was the same as that used in the delamination tests. An artificial notch was deliberately created at the edge of the film with a 3-mm strip. Epoxy resin was applied to the upper and lower surfaces of the specimens to ensure that they adhered to the stage. A force of 100 N was applied to the sample for 30 s to enhance adhesion. Subsequently, the sample was fixed onto the tensile test fixture. Before obtaining the stress versus displacement curve, the samples were manually pre-tensioned to 5 N. The stretching rate was controlled manually and set to a final rate of 0.01 mm s^{-1} for a general perovskite sample, ensuring repeatability in the results. The experiment was terminated when the lower and upper glass sections detached or when the load had decreased to less than 1 N. We used the following formula to calculate the traction stress:

$$S = \frac{F}{A} \tag{1}$$

where F is the applied load (force magnitude) and A is the area of the specimen.

For the adhesion toughness measurements, the sample structure was the same as that used for the delamination test except for the area (37.5 mm × 12.5 mm). A notch was created with a 5 mm × 12.5 mm strip at the short edge of the sample. Before mechanical testing, artificial plane cracks were made by inserting a blade into the notch. The sample stage was glued to both sides of the glass substrate at the crack end. The toughness was measured with the double-cantilever-beam test described in the literature[24]. The toughness G_c was calculated with the following equation:

$$G_c = \frac{12P_c^2a^2}{B^2E'h^3}\left(1 + 0.64\frac{h}{a}\right)^2, \tag{2}$$

where P_c is the critical load, a is the crack length, B is the width (12.5 mm), E' is the plane strain modulus of the substrate and h is the half-height of the sample (1 mm). E' is given by

$$E' = \frac{E}{1-\upsilon^2}, \tag{3}$$

where E is the Young's modulus for the substrate (70 GPa) and υ is Poisson's ratio for the substrate (0.35). The crack length was calculated using the compliance relation:

$$a = \left(\frac{\mathrm{d}\Delta}{\mathrm{d}P}\frac{BE'h^3}{8}\right)^{\frac{1}{3}} - 0.64h \tag{4}$$

where Δ is the displacement and P is the load.

Device fabrication and testing

We used the following procedure to fabricate the n–i–p devices. An ITO substrate with an anti-reflection layer was cleaned by sonication for 30 min in detergent, deionized water, acetone and IPA, respectively, and then subjected to a 30-min ultraviolet/ozone treatment. For the TIE devices, 1 mg AM was added to 1 ml SnO_2 colloidal solution, which was then deposited onto the substrate by spin-coating at 3,000 rpm over 30 s, followed by annealing at 100 °C for 5 min. The perovskite precursor of $(FA_{0.95}Cs_{0.05}PbI_3)_{0.975}(MAPbBr_3)_{0.025}$ was prepared by mixing 228.8 mg FAI, 33.67 mg MACl, 18.2 mg CsI, 18.2 mg $MAPbBr_3$ and 705.3 mg PbI_2 in 1 ml DMF/DMSO mixed solvent (9:1, v/v). For the TIE device, 1 mg TMTA and 0.1 mg AIBN were added to 1 ml perovskite precursor solution. The perovskite precursor was deposited onto an ITO/SnO_2 substrate by two consecutive spin-coating steps of 1,000 rpm for 10 s and 4,000 rpm for 30 s. During the last 10 s, 800 µl diethyl ether was dripped onto the centre of the substrate, followed by annealing at 100 °C for 40 min. For later passivation, a solution of 3 mg ml^{-1} 4-OMe-PEAI in IPA was deposited onto the perovskite surface at a spin rate of 4,000 rpm over 30 s. The hole-transport layer was deposited onto the passivation layer at a spin rate of 3,000 rpm over 30 s, which consisted of 90 mg spiro-OMeTAD, 21 µl Li-TFSI (520 mg in 1 ml acetonitrile), 16 µl FK209-Co(III)-TFSI (375 mg in 1 ml acetonitrile) and 36 µl 4-tert-butylpyridine in 1 ml CB. For the device stability test, PTAA was used instead of spiro-OMeTAD as the hole-transporting layer. The solution was prepared by mixing 30 mg of PTAA and 3 mg of TPFB in 1 ml CB. The PTAA was deposited onto the passivation layer at a spin rate of 3,000 rpm over 30 s. Finally, an Au electrode with a thickness of 80 nm and an active area of 0.09 cm^2 was deposited by thermal evaporation.

We used the following procedure to fabricate the p–i–n devices. The NiO_x nanoparticles (15 mg ml^{-1} in distilled water) were spin-coated onto an ITO substrate with an antireflection layer at 3,000 rpm for 40 s and annealed at 100 °C for 5 min in ambient air. For the TIE devices, 1 mg AM was added to 1 ml NiO_x solution. For the preparation of $Cs_{0.05}(MA_{0.05}FA_{0.95})_{0.95}Pb(I_{0.95}Br_{0.05})_3$ perovskite layers, a 1.5 M precursor solution was prepared by dissolving PbI_2 (676.8 mg), FAI (232.8 mg), CsI (19.5 mg), $PbBr_2$ (41.3 mg) and MACl (35.2 mg) in 1 ml DMF/DMSO mixed solvent (4:1, v/v). Then, 30 µl Me-4PACZ (2.4 mg in 1 ml DMF) was added to the perovskite precursor solution. For the TIE devices, 1 mg TMTA and 0.1 mg AIBN were added to 1 ml perovskite precursor solution. The resultant perovskite precursor solution was then spin-coated at 4,000 rpm over 40 s using 200 µl of ethyl acetate as the antisolvent at the 30th second. The film was annealed at 120 °C for 20 min. After coating PEAI (3 mg ml^{-1} in IPA) onto the perovskite film at 4,000 rpm over 20 s, the $PC_{61}M$ layer (20 mg ml^{-1} in CB) and the bathocuproine film (0.7 mg ml^{-1} in IPA) were spin-coated at 2,000 rpm and 4,000 rpm over 20 s, respectively, and then annealed at 70 °C for 10 min. Finally, a 100-nm-thick layer of Ag was evaporated onto the active area (0.09 cm^2) through a mask.

For current density–voltage (J–V) characterization of PSCs, measurements were conducted with a source meter (2400, Keithley) under

AM 1.5G primary solar illumination (100 mW cm^{-2}) produced by a solar simulator (Sirius SS, Zolix). The scan voltage range extended from −0.1 V to 1.23 V in opposite directions with a step size of 0.02 V and a dwell time of 10 ms. The light intensity was calibrated using a silicon-based reference solar cell. The reference cell used to calibrate the solar stimulator before the *J*–*V* measurements was an Oriel reference solar cell accredited to the ISO-17025 standard developed by the National Institute of Standards and Technology. The area of the metal aperture used during the test was 0.069 cm^2, as certified by the National Institute of Metrology, China (Report Number CDjc2024-06131). The EQE was recorded at a chopping frequency of 165 Hz in a.c. mode on a solar cell quantum efficiency measurement system (QE-R3011, Enlitech).

For the stability tests, we used n–i–p PSCs with the ITO/SnO_2/perovskite/PTAA/Au configuration. For the damp heat test, the PSCs were all encapsulated with an ultraviolet-curing adhesive and a cover glass. The edge of the cover glass was protected by epoxy and cured at RT for >48 h. The encapsulated devices were placed in an environmental chamber at 85 °C and 85% relative humidity, and PCEs were measured periodically after the devices had cooled to RT. Experiments were conducted to evaluate the long-term operational stability of the unencapsulated devices under a flowing nitrogen atmosphere. The stability test involved tracking the MPP under continuous illumination from a white LED lamp at an intensity of 100 mW cm^{-2}. Additionally, the MPP tracking stability at 85 °C was assessed using a built-in heating stage. For the thermal-cycling stability test (following the ISOS-T-3 protocol), encapsulated devices were placed in an environmental chamber programmed to cycle the temperature between −40 °C and 85 °C, with each complete cycle lasting 6 h.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data that support the main findings are available in the main text and Supplementary Information. Source data are provided with this paper.

References

51. Shi, D. et al. Low trap-state density and long carrier diffusion in organolead trihalide perovskite single crystals. *Science* **347**, 519–522 (2015).
52. Zhong, J. et al. Room temperature fabrication of SnO_2 electrodes enabling barrier-free electron extraction for efficient flexible perovskite photovoltaics. *Adv. Funct. Mater.* **32**, 2200817 (2022).

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Author contributions

Y.Z. conceived of the idea and directed the project. Y.Z., W.Y. and P.G. designed the experiments. P.G. and W.Y. co-performed the film characterizations and property measurements. W.Y. fabricated the devices. P.G. and W.Y. tested the devices. K.W. and S.L. performed the TEM characterization. M.H. contributed to the conceptual illustration. T.D., W.Y. and M.H. contributed to the mechanical tests. Y.Z., P.G. and W.Y. prepared the paper. All co-authors contributed to the review and revision of the article.

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Competing interests

A US patent was filed based on this work by Y.Z., P.G. and W.Y. The other authors declare no competing interests.

Additional information

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Solar Cells Reporting Summary

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Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells	☒ Yes	We provided the information in the Methods. The active area is 0.09 cm2.
	☐ No	*Explain why this information is not reported/not relevant.*
Method used to determine the device area	☒ Yes	The active area was determined by the shading mask and further calculated by an optical microscope.
	☐ No	*Explain why this information is not reported/not relevant.*

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes	We provided the J-V plots in both directions in Supplementary Fig. 35 and Supplementary Fig. 44.
	☐ No	
Voltage scan conditions	☒ Yes	J-V curves were measured by reverse scan (1.23 V to -0.01 V) and forward scan (-0.01 V to 1.23 V) with a scanning rate of 200 mV/s (voltage step of 20 mV).
	☐ No	*Explain why this information is not reported/not relevant.*
Test environment	☒ Yes	Performance measurements were carried out in a nitrogen environment.
	☐ No	*Explain why this information is not reported/not relevant.*
Protocol for preconditioning of the device before its characterization	☐ Yes	*Provide a description of the protocol.*
	☒ No	There is no protocol for preconditioning of the device before its characterization.
Stability of the J-V characteristic	☒ Yes	Stabilized maximum power output characterizations were conducted (Fig. 3c, Supplementary Figs. 45-47).
	☐ No	*Explain why this information is not reported/not relevant.*

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☒ Yes	The target cells showed negligible hysteresis .
	☐ No	*Explain why this information is not reported/not relevant.*
Related experimental data	☒ Yes	We provided these data in Supplementary Fig. 35 and Supplementary Fig. 44
	☐ No	*Explain why this information is not reported/not relevant.*

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes	We provided the information in Fig. 3b and Supplementary Fig. 37.
	☐ No	*Explain why this information is not reported/not relevant.*
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes	We provided the information in Fig. 3b. The integrated Jsc from EQE spectra is consistent with the Jsc from J-V measurements.
	☐ No	*Explain why this information is not reported/not relevant.*

For tandem solar cells, the bias illumination and bias voltage used for each subcell

☐ Yes
☒ No

Provide a description of the measurement conditions.

There is no tandem solar cell in this research.

5. Calibration

Light source and reference cell or sensor used for the characterization

☒ Yes
☐ No

We provided the information in Methods.

Explain why this information is not reported/not relevant.

Confirmation that the reference cell was calibrated and certified

☒ Yes
☐ No

We provided the information in Methods.

Explain why this information is not reported/not relevant.

Calculation of spectral mismatch between the reference cell and the devices under test

☐ Yes
☒ No

Provide a value of the spectral mismatch and/or a description of how it has been taken into account in the measurements.

We did not calculate the spectral mismatch between the reference cell and the devices under test because our current level of solar cell performance measurements is sufficient for verifying the efficiency improvement based on our new method

6. Mask/aperture

Size of the mask/aperture used during testing

☒ Yes
☐ No

The area of the mask used during testing is 0.069 cm2.

Explain why this information is not reported/not relevant.

Variation of the measured short-circuit current density with the mask/aperture area

☐ Yes
☒ No

Report the difference in the short-circuit current density values measured with the mask and aperture area.

We did not study the variation of short-circuit current density as function of the mask area, because this is not relevant to the scope of this study.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☒ Yes
☐ No

We performed the PCE certification in an independent certification center (Tianjin Institute of Metrological Supervision and Testing, China) with the CNAS accreditation.

Explain why this information is not reported/not relevant.

A copy of any certificate(s)

☒ Yes
☐ No

We provided the PCE certification report in Supplementary Fig. 34.

Explain why this information is not reported/not relevant.

8. Statistics

Number of solar cells tested

☒ Yes
☐ No

We provided the information in Supplementary Fig. 36. 50 individual devices under each condition were tested.

Explain why this information is not reported/not relevant.

Statistical analysis of the device performance

☒ Yes
☐ No

We provided the information.

Explain why this information is not reported/not relevant.

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions

☒ Yes
☐ No

We provided the information in Fig. 3c, Fig. 3d, Supplementary Figs. 45-47, Supplementary Fig. 50, and Supplementary Fig. 51 . We followed the ISOS-T-3, ISOS-L-1, ISOS-L-2, and ISOS-D-3 protocols to test the long-term stability.

Explain why this information is not reported/not relevant.